

1. INTRODUCTION

This project aims to design and implement an **ATM Simulation System** using **8086 Assembly Language**. The system will allow a user to log in using a PIN, check their balance, withdraw funds, deposit funds, and exit safely. The project demonstrates key concepts of Assembly programming, including:

- Register usage (AX, BX, CX, DX)
- Loops & condition checks
- Modular programming using procedures
- Interrupts (INT 21h) for input/output
- Input validation for user safety
- Control flow and branching

The ATM Simulation project helps students apply COAL concepts in a practical, real-world scenario, allowing deeper understanding of low-level programming and computer architecture.

2. OBJECTIVES

The primary objectives of the ATM Simulation Project are:

- To simulate real ATM operations using 8086 Assembly Language.
- To implement secure PIN verification with limited attempts.
- To allow user-triggered financial operations (balance inquiry, withdrawal, deposit).
- To practice modularity by dividing the program into reusable procedures.
- To build a safe and user-friendly interface through interrupt-based I/O.
- To strengthen understanding of registers, loops, jumps, and stack operations.

3. FUNCTIONAL REQUIREMENTS

The ATM program will include:

✓ PIN Verification

- User enters a 4-digit PIN.
- Incorrect PIN decreases attempts.
- After 3 failed attempts, system locks and exits.

✓ Main Menu Options

1. Check Balance
2. Withdraw Cash
3. Deposit Cash
4. Exit

✓ Program Features

- Validate numeric input
- Prevent negative or invalid entries
- Deny withdrawals greater than balance
- Clear and readable interface
- Efficient execution compatible with EMU8086 / TASM

4. ALGORITHM

Start

1. Initialize balance (e.g., 5000).
2. Display welcome message.
3. Ask user for 4-digit PIN.
4. Compare entered PIN:
 - If correct → go to main menu
 - If wrong → reduce attempts
 - If attempts = 0 → lock system and exit

Main Menu

1. Display menu options:
 - 1 – Check Balance
 - 2 – Withdraw
 - 3 – Deposit
 - 4 – Exit
2. Read user choice.

3. Execute selected procedure.
4. Return to menu until user exits.

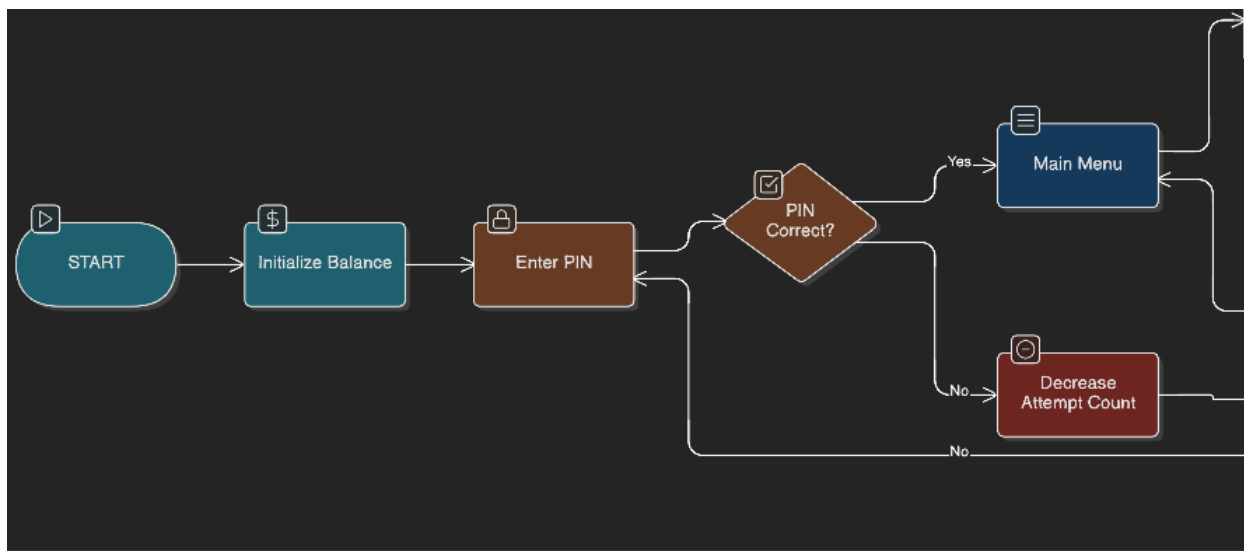
Withdrawal Procedure

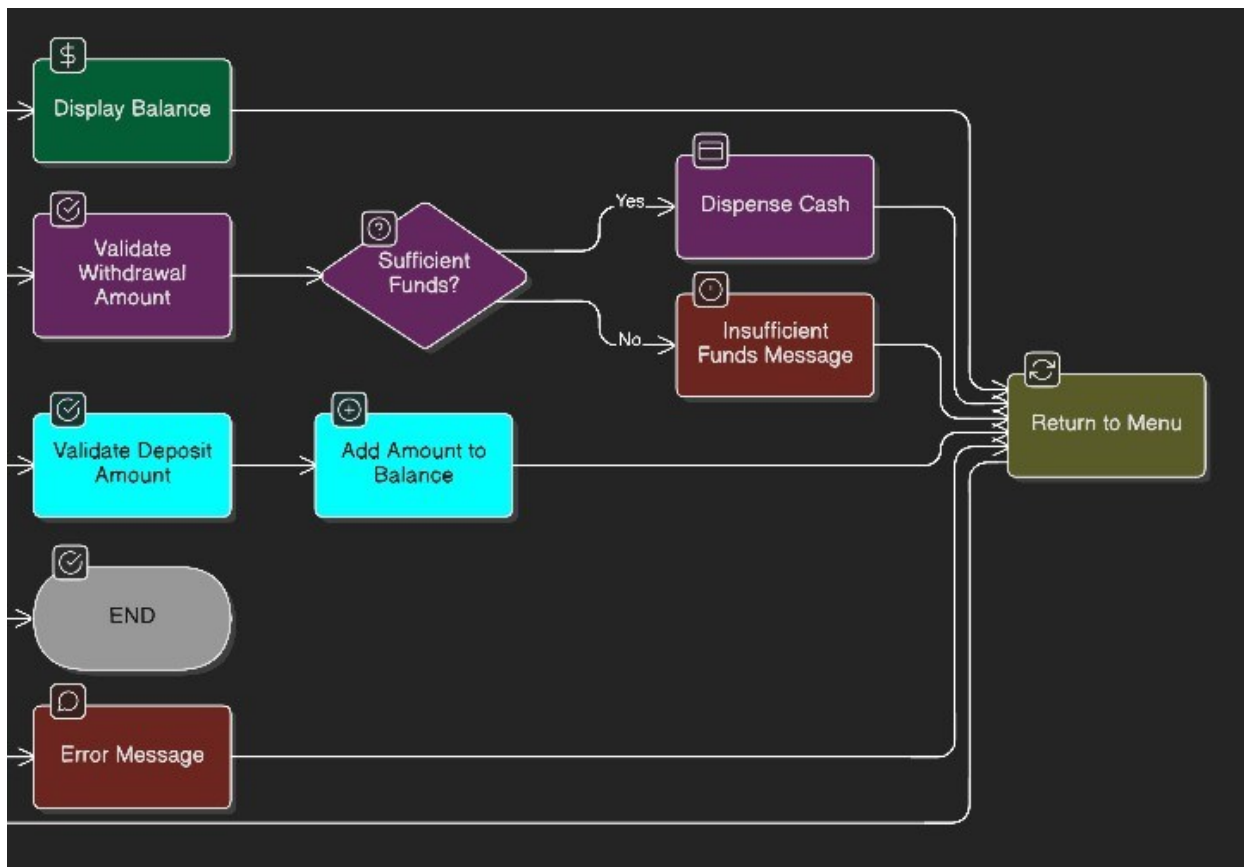
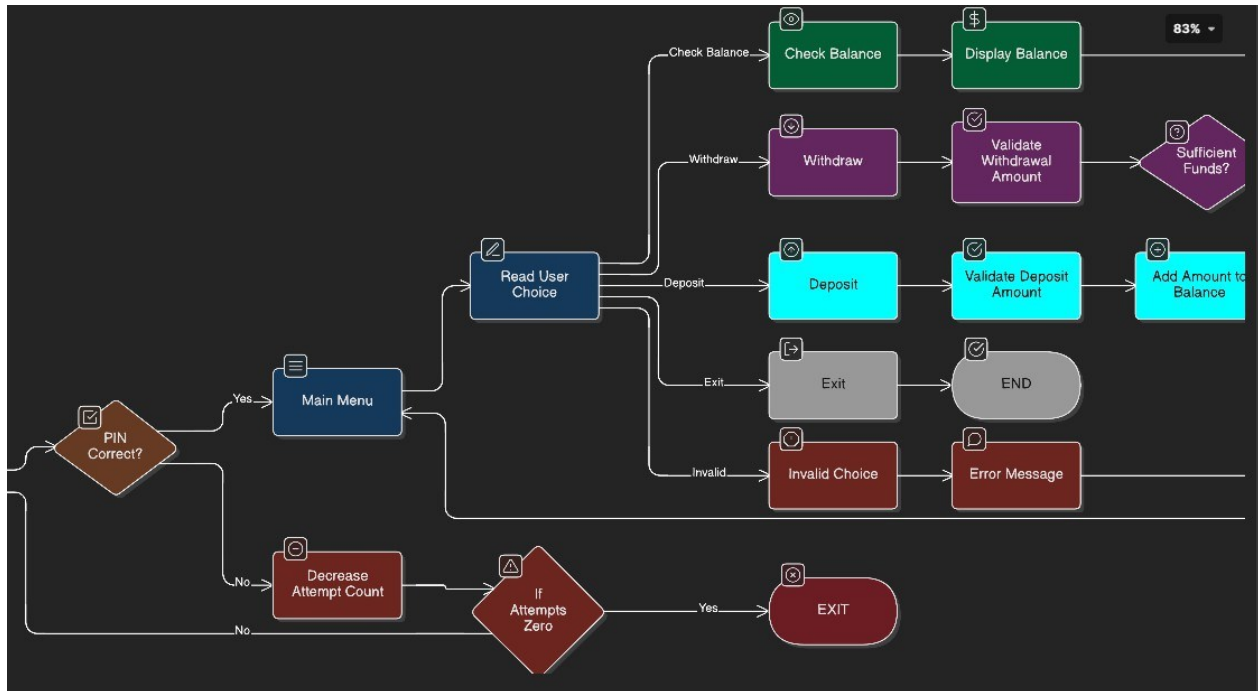
1. Ask user to enter withdrawal amount
2. Validate input
3. If amount > balance → show error
4. Else subtract from balance and confirm

Deposit Procedure

1. Ask user to enter deposit amount
2. Validate input
3. Add deposit to balance
4. Display success message

End Program





EXPECTED RESULTS & CONCLUSION

The ATM Simulation Program will successfully demonstrate:

- Effective use of 8086 registers
- Proper input/output handling using interrupts
- Logical flow using conditional jumps and loops
- Modular code organization using procedures
- Realistic ATM functionality within a low-level environment

This project will not only strengthen our understanding of Assembly Language but also highlight how low-level programming forms the foundation of real-world computational systems.